Bhukya Bharath

ee25mtech11029@iith.ac.in | bvknaik4220@gmail.com | +91 7989744063 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.04
SRI NARAYANA JUNIOR COLLEGE	Class XII (BOARD OF INTERMEDIATE EDUCATION ANDHRA PRADESH)	2014	80.30%
A P T W R SCHOOL BOYS MARIPEDA	Class X (BOARD OF SECONDARY EDUCATION ANDHRA PRADESH)	2012	7.20%

INTERNSHIPS

Student Intern - Electrical Maintenance Division (O&M)

Kothagudem Thermal Power Station (KTPS - A Station), **TSGENCO**

- Completed 4 weeks of hands-on industrial training under Divisional Engineers, actively monitoring **switchgear operations** and protection schemes for utility turbo-generators and generator-transformer units.
- Traced control panel wiring and verified trip logic for **percentage differential protection**, stator earth-fault circuits, and negative phase sequence filters during routine maintenance.
- Examined single-line diagrams, operating coils, and contact timing for **Air and SF6 circuit breakers**, reviewing breaker failure backup protection under live plant conditions.

PROJECTS

M.Tech Thesis

Achieving Consensus for Ground Robots in Practice

Hardware & Tools: ROS 2 Jazzy, TurtleBot 3 Burger, TurtleBot 4, Python, MATLAB, GPOPS-II

- **Modeled** non-holonomic mobile robot motion by studying existing kinematic and dynamic equations for a **TurtleBot 3 Burger**, mapping wheel velocities to state trajectories in MATLAB.
- **Formulated** point-to-point path planning using **optimal control theory**, generating minimum-time and minimum-energy control profiles under practical actuator bounds.
- **Solved** constrained numerical trajectory optimization problems using **GPOPS-II**, generating feasible state and input profiles between coordinate waypoints.
- **Built** the execution and communication setup in **ROS 2** to test planned trajectories on hardware and extend the solver toward **multi-robot decentralized consensus**.

M.Tech Advanced Control Lab Experiments

Hardware & Tools: MATLAB/Simulink, Quanser QUBE-Servo, 3-DOF Helicopter, Inverted Pendulum, Ball & Beam

- Worked with real-time nonlinear and unstable experimental setups, including the **3-DOF Helicopter**, **Linear Inverted Pendulum**, **Ball and Beam**, and **QUBE-Servo**.
- Derived **state-space models** from physical differential equations using linearization methods, and designed state-feedback and estimation controllers, including **Pole Placement**, **LQR**, **Kalman Filter**, and **LQG**.
- Implemented **PV and cascaded PD controllers** on these test setups and evaluated their control performance.
- Compared simulation and hardware experimental results, using error values to analyze the differences between

theoretical and real-time responses.

Quantized Predictor-Based Control for Time-Delay Systems

MATLAB -- Backstepping -- Delay Compensation -- Quantized Feedback

- Implemented predictor-feedback control laws in **MATLAB** to resolve state trajectory instability on **2-DOF helicopter** and **3-DOF hovercraft** models.
- Formulated a **backstepping predictor controller** to compensate for a **0.2 s actuator delay** while preserving asymptotic stability against signal quantization steps.
- Analyzed closed-loop convergence rates and transient performance under state and input quantization constraints through numerical simulations.

B.Tech Projects

Superconducting Fault Current Limiter in Electrical Power Systems **(Major Project)**

Protection of Generator Using Relays and Circuit Breakers **(Mini Project)**

CERTIFICATIONS

- **Certificate of Appreciation in MATLAB** - Awarded by **Capricot Technologies** for core modeling and simulation proficiency.
- **Industrial Switchgear & Protection Training** - Practical training on switchgear testing and relay operation conducted by **Schneider Electric & LS Electric**.
- **ROS 2 Developer Training (Jazzy Jalisco)** *(Ongoing)* - Practical coursework covering **URDF, TF2 transforms, Gazebo physics simulation, and custom nodes**, Udemy.
- **Python & Machine Learning Foundations** - Verified specialization covering **data processing, regression, and model validation**, Kaggle.

AREA OF EXPERTISE

- **Core Control & Dynamics: State-Space Modeling**, Linear Quadratic Regulator (**LQR**), LQG, **Optimal Control**, Non-linear Control, Backstepping, Lyapunov Stability, **Kalman Filter**, System Identification, Differential Riccati Equations
- **Software & Toolkits: MATLAB/Simulink**, SimPowerSystems, **ROS 2 (Jazzy)**, Linux, Gazebo, RViz, Git/GitHub, LaTeX
- **Lab Hardware & Platforms: TurtleBot 3 & 4**, Quanser IP02 Linear Cart, Quanser 3-DOF Hovercraft, **Quanser BB01 Ball & Beam**
- **Optimization & Programming: Convex Optimization** (SOCP, QP), Dynamic Programming, **Python**, C (Basics)

POSITION OF RESPONSIBILITIES

Teaching Assistant - Control Systems & Optimal Control

- Guided undergraduate students in **Control Systems** and postgraduate students in **Optimal Control** through weekly tutorial problem-solving sessions, coursework assignments, and theoretical clarifications under the guidance of the faculty instructor.
- Assisted the course instructor in organizing mid-semester and end-semester examinations, actively handling **invigilation duties and evaluating answer papers** for both undergraduate and postgraduate batches.

Core Member - Akshara Sanjeevini (Telugu Cultural Club)

- Coordinated core-team planning for the club's official inauguration; organized event schedules, handled guest hospitality, and facilitated the **official name and logo launch** during an invited Ashtavadhani scholar's campus visit.

Core Team Coordinator - Prayas, Rural Development Centre (RDC)

- Leading student volunteer teams to plan and execute **educational outreach drives and community initiatives** in nearby rural villages across the Sangareddy district.

Volunteer - Prayas, Rural Development Centre (RDC)

- Supported field-level activities and weekend teaching programs, helping coordinate **science demonstrations and basic learning workshops** for government school students.

RELEVANT COURSES

- **M.Tech Coursework: Optimal Control**, Introduction to State Space, **State Estimation**, Matrix Theory, Random Variables & Stochastic Processes, **Advanced Control Lab**, Convex Optimization, Automotive Communication & Sensing, Smart Grid, Memory Circuits & System Design, **Multi-Agent Systems**, Machine Learning and Its Applications, **Fundamentals of Robotics**, Energy Management

HOBBIES & INTERESTS

- **Sports & Games:** Playing **Cricket**, **Carrom**, and **Chess** during weekends and institute inter-hostel matches.
- **Agriculture & Farming:** Supporting family farming operations during semester breaks in cultivating **paddy, cotton, and chilli crops** in my hometown.
- **Cinema:** Watching **Spy, Biography, Classic, and Action films** in theatres and on OTT platforms.
- **Cooking:** Cooking traditional recipes like **Hyderabadi Biryani** and **Paneer Butter Masala** for friends and family.
- **Social Volunteering:** Guiding students from **Rural Government Schools and Colleges** on higher education awareness, engineering admissions, and **GATE exam preparation**.